FOR A PROPER GEOMETRICALLY NORMAL SCHEME: TRIVIAL GLOBAL FUNCTIONS, GEOMETRIC CONNECTEDNESS, IRREDUCIBILITY AND INTEGRALITY ARE EQUIVALENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and pairs off the four conditions in three steps; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-normal-stein is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The Stein type equivalence between trivial global functions and geometric connectedness for a proper geometrically reduced scheme. Same chapter, Lemma lemma-geometrically-reduced-stein, invoked at proof lines 2282-2283. Supplies the equivalence of the first two conditions.
- A locally Noetherian normal scheme is a disjoint union of its irreducible components. Properties, Lemma properties-lemma-normal-Noetherian, invoked at proof lines 2283-2286. Supplies the equivalence between connectedness and irreducibility after base change.
- Reducedness upgrades irreducibility to integrality. Standard, used at proof lines 2288-2289. Supplies the equivalence of irreducibility and integrality for the base change.

Source reference. Original lemma 0FD2: sources/stacks-varieties/pinned-original-source.tex, source lines 2075–2075. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 033M: sources/stacks-chow/tags.txt, source lines 4061–4061. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a proper geometrically normal scheme over k. The following are equivalent*

(1) $H^0(X, \mathcal{O}_X) = k$,

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(2) *X is geometrically connected,*
(3) *X is geometrically irreducible, and*
(4) *X is geometrically integral.*

Proof. By Lemma 0FD2 we have the equivalence of (1) and (2). A locally Noetherian normal scheme (such as $X_{\overline{k}}$) is a disjoint union of its irreducible components (Properties, Lemma 033M). Thus we see that (2) and (3) are equivalent. Since $X_{\overline{k}}$ is assumed reduced, we see that (3) and (4) are equivalent too. □